Team Control Number

12704

Problem Chosen

C

2022

HiMCM/MidMCM

Summary Sheet

Summary

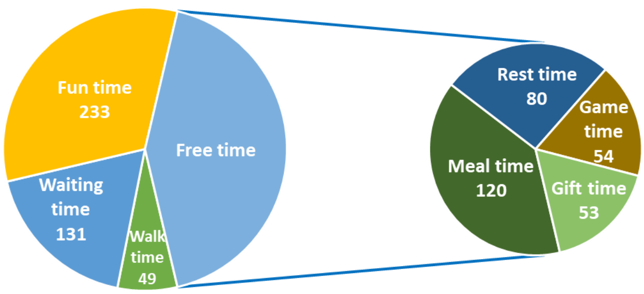
A group of four friends has hired our team at the MidMCM Travel Agency to make sure each member has a great time during their one-day trip to Polygon Paradise Park. Our team is to develop a schedule to make the group's day at the park the best possible.

There are 17 different locations in this park, we call them as X_i. First, we assign these 17 locations coordinate values in pixels, and use the Euclidean distance to calculate the pixel distance between any two locations. Then, we convert the pixel distance to walking time, and find the center location is X_{13}(Restroom, WC). Third, we transform the problem into a TSP problem, and find the shortest traversal path. Fourth, on the basis of the shortest traversal path, we divide the park map into 4 parts. Before the partitioning, 37% of walk times were 5 minutes or more. After partitioning, the walking time of the farthest two points is 5 minutes.

When develops the schedule, we arrange key locations first, and divide the 12 hours into four different sections, each section corresponds to a part of the park. After completing the development of the schedule, we gave each member of the team corresponding notes.

There are five highlights in our schedule.

- The average waiting time value for completing 10 rides only 7.5 minutes.
- The longest walking time not exceed 5 minutes.
- The schedule includes enough free time.
- Fully meet everyone's different needs.
- All rides and shows will be completed.



The time allocation arrangement in one day

Contents

1. A letter to my friends

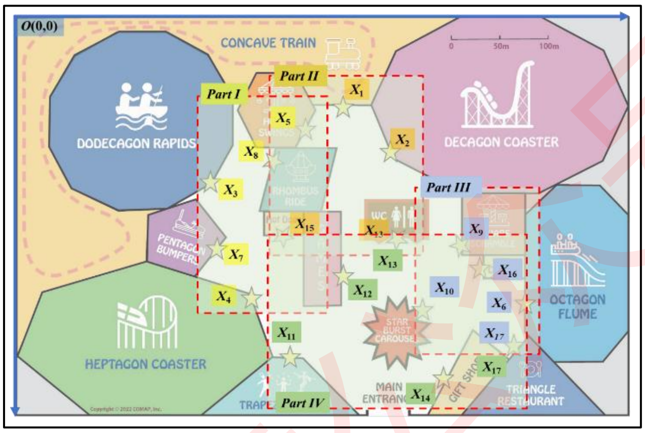
Dear Ming、Ishmael、Karine、Freya，

Thanks for hiring our team at the MidMCM Travel Agency to develop a schedule for your one-day trip to the Polygon Paradise Park. We will try our best to provide you an ideal day guide at the park. Let’s Familiarize with Polygon Paradise Park at first.

Introduction to the Park

Polygon Paradise Park is a small amusement park open from 9am to 9pm. The park has ten rides of different types and thrill levels. The Trapezoid Show, a location with several different performances throughout the day, offers a circus acrobatics show, a magic show, and evening fireworks. The Games building has both arcade and carnival games. Food options include the Triangle Restaurant, the hot dog stand, and the ice cream cart. The hot dog stand and the ice cream cart are “take away” options, while the Triangle Restaurant requires 30-60 minutes to sit at a table and enjoy a meal. The restrooms (WC) are centrally located in the park for easy access. Visitors can purchase souvenirs in the gift shop located near the park entrance.

Ideal day guide at Polygon Paradise Park



Map of the Park

Part	Start Time	Location	Wait Time	Time taken	Next Location	Walking Time	End Time	Time	Tips
Part I 09:00 -10:00	9:00	X4	0:10	0:02	X7	0:02	9:14	09:00-10:00	Heptagon Coaster
	9:14	X7	0:00	0:15	X3	0:02	9:31	09:00-10:00	Pentagon Bumpers
	9:31	X3	0:00	0:04	X8	0:02	9:37	09:00-10:00	Dodecagon Rapids
	9:37	X8	0:05	0:03	X5	0:01	9:46	09:00-10:00	Rhombus Ride
	9:46	X5	0:00	0:05	X1	0:01	9:52	09:00-10:00	Hexa-Swings
Part II 10:00 -11:00	9:52	X1	0:05	0:03	X13	0:03	10:03	09:00-10:00	Concave Train
	10:03	X2	0:30	0:03	X15	0:03	10:39	10:00-11:00	Decagon Coaster
	10:39	X15	0:05	0:07	X13	0:03	10:54	10:00-11:00	Hot Dog
	10:54	X13	0:00	0:06	X9	0:02	11:02	10:00-11:00	WC
Part III 11:00 -14:00	11:02	X9	0:10	0:10	X16	0:01	11:23	11:00-12:00	Square Scramble
	11:23	X16	0:05	0:25	X6	0:02	11:55	11:00-12:00	Icecream
	11:55	X6	0:10	0:03	X17	0:01	12:09	11:00-12:00	Octagon Flume
	12:09	X17	0:30	1:00	X10	0:02	13:41	12:00-13:00	Lunch
	13:41	X10	0:05	0:05	X11	0:03	13:54	13:00-14:00	Star Burst Carousel
Part IV 14:00 -21:00	13:54	X11	0:06	1:00	X12	0:03	15:03	14:00-15:00	Magic Show
	15:03	X12	0:00	0:54	X11	0:03	16:00	15:00-16:00	Games
	16:00	X11	0:00	1:00	X14	0:05	17:05	16:00-17:00	Circus Acrobatics
	17:05	X14	0:00	0:53	X17	0:02	18:00	17:00-18:00	Gift Shop
	18:00	X17	0:10	1:00	X13	0:04	19:14	18:00-19:00	Dinner
	19:14	X13	0:00	0:42	X11	0:04	20:00	19:00-20:00	WC
	20:00	X11	0:00	1:00	/	0:00	21:00	20:00-21:00	Firework Show

Schedule of the one-day trip

Now let's take a look at the schedule of this trip.

- First, this park will be divided into 4 parts and including 17 locations in our schedule. You will start the one-day trip at the location of Heptagon Coaster. Please arrive at this location at 9am on time. You will play in the park clockwise to minimize the waiting time and walking time.
- You will have lunch about 12:15, and you can buy hot dogs as snacks. Considering Karine doesn't like hot dogs very much, we suggest that she can bring her own snacks.
- After lunch, most of the queues in all attractions are already more than 30minutes. The afternoon and the late afternoon are the time for shows, games, mini-attraction, or some photo opportunities in the garden.

- You will have dinner about 18:00, After your stomach is feeling good and you are feeling good, you can stick around and watch the fireworks show.
- After the fireworks show, it's time to say goodbye. We have completed the exploration of all projects. I hope you are satisfied with this schedule.

Highlights of the Schedule

There are five highlights in our schedule.

- Arrange the right project at the right time to reduce the waiting time. According to our schedule, the average waiting time for completing all 10 rides only 10 minutes.
- During this day's trip, time will not waste on walking. The total walking time will be less than 1 hour. The longest walking time shall not exceed 5 minutes.
- The schedule includes enough leisure time, which can be used to have a meal, take a rest, play games or buy souvenirs.
- At the same time, we will also make targeted arrangements according to the different needs of everyone to make the collective satisfaction as high as possible.
- All rides and shows will be completed, and individual rides can be tried more than once.

Four tips

It is very important to know the following tips.

- Pack light and wear sneakers. Before you go, think about what you absolutely need to bring with you. There are lockers at every roller coaster to drop your bag, but it might slow you down. And don't wear flip flops. On rides like Decagon Coaster, you will have to take them off. Plus, sneakers will help you get from one location to the other as fast as possible.
- Follow the rules, not the other kids. The rules are to help you have fun, not spoil it.
- Pay attention to specific safety instructions. Watch out when the rules say things like "hang on to the handles," "slide only feet first," "stay seated," "don't rock the seat," "get rid of gum before you ride" or "no flipping." When they post those rules, it is because they know those actions can cause kids to get hurt.
- Read the signs and teach your parents some safety rules. Even adults get hurt when they put their hands in the air or don't ride properly.

The final last, buy yourself a t-shirt to celebrate and a souvenir of your accomplishment! I hope you'll have fun and make the best out of your time.

Sincerely yours，

Team #12704

2. Problem Restatement

A group of four friends has hired our team at the MidMCM Travel Agency to make sure each member has a great time during their one-day trip to Polygon Paradise Park. Our team is to develop a schedule to make the group's day at the park the best possible. We have following information.

About The Park

Polygon Paradise Park is a small amusement park open from 9am to 9pm. The park has ten rides of different types and thrill levels. The Trapezoid Show, offers a circus acrobatics show, a magic show, and evening fireworks. The Games building has both arcade and carnival games. Food options include the Triangle Restaurant, the hot dog stand, and the ice cream cart. The hot dog stand and the ice cream cart are "take away" options, while the Triangle Restaurant requires 30-60 minutes to sit at a table and enjoy a meal. The restrooms (WC) are centrally located in the park for easy access. Visitors can purchase souvenirs in the gift shop located near the park entrance.

About The Group

Name	Like	Dislike
Ming	play games and win prizes	fast moving rides (especially after eating)
Ishmael	roller coasters、magic show	
Karine	circus acrobatics show	waiting in long lines、hot dogs
Freya	water rides、with friends、souvenir	

About The Planning Resources

Our planning resources from MidMCM Travel Agency include a show schedule for the Trapezoid Show location, tables containing ride categories and durations, expected ride wait times throughout the day, food options and wait times, and a map of the park. These resources are included at the end of the problem statement.

What We Need to Do

- **Get Started.** Familiarize with Polygon Paradise Park and the group of friends.
- **Create the Schedule.** Create a detailed schedule for the group of friends to use as a guide for their day at Polygon Paradise Park.
- **Share Our Schedule.** Write a one- to two-page letter to the group of friends that describes the highlights of our recommended schedule and explains why it will offer the best possible day at Polygon Paradise Park.
- **Reflect.** Try to use the process we used to create the schedule for Ming, Ishmael, Karine, and Freya to create a schedule for our team, and briefly describe the process and what would be different!

3. Problem Assumptions

- **Assumption 1**: The distance between any two points is a straight-line distance, and the round-trip distance is equal.
- **Assumption 2**: The playing time and waiting time of each ride are consistent with the data given in the table.
- **Assumption 3**: The play process of all rides is smooth, without considering unexpected situations.
- **Assumption 4**: The schedule can be executed correctly
- **Assumption 5**: Walking speed is calculated as 45 m/s

4. Variable Definitions

A table of variables used in the study is given below.

Notation	Definition
$X_i \ (i=1,2,\cdots,17)$	Different location
$(x_i, y_i) \ (i=1,2,\cdots,17)$	coordinate of X_i
$D_{ij} \ (i,j=1,2,\cdots,17)$	Distance between location i and location j
$Tw_{ij} \ (i,j=1,2,\cdots,17)$	Time required to walk from location i to location j
$Tave_i \ (i=1,2,\cdots 17)$	Average time from each location to others
$\lfloor \ \rfloor$	Round up the value to the nearest integer
$\min \ T=\sum_{i=1}^{17}\sum_{i=1}^{17}\left(Tw_{ij}\right)\times x_{ij}$	The minimum value of the total distance function
$\sum_{i\neq j,i=1}^{n} x_{ij}=1$	Only one point at a time
$\sum_{i\neq j,j=1}^{n} x_{ij}=1$	Each point is passed once

5. Model Description

5.1 Establish coordinate system

There are 17 different locations in this park, we call them as $X_i(i=1,2,\cdots 17)$. The corresponding notation is shown in Table 1.

Table 1 Names of each location

Type	Location	Notation	Type	Location	Notation
Rides	Concave Train	X_1	Others	Trapezoid Show	X_{11}
	Decagon Coaster	X_2		Games	X_{12}
	Dodecagon Rapids	X_3		WC	X_{13}
	Heptagon Coaster	X_4		Gift Shop	X_{14}
	Hexa-Swings	X_5		Hot Dog Stand	X_{15}
	Octagon Flume	X_6		Ice Cream Cart	X_{16}
	Pentagon Bumpers	X_7		Triangle Restaurant	X_{17}
	Rhombus Ride	X_8			
	Square Scramble	X_9			
	Star Burst Carousel	X_{10}			

To facilitate the calculation of the direct distance between any two locations, we first establish a coordinate system based on the park map. In our coordinate system, the upper left corner is the coordinate origin, the right direction is the X-axis, the downward direction is the Y-axis. The size of the whole park map is 1172×1146 pixels, we take one pixel as a unit. So, we can get the coordinate value of each location by a free software which is called "GetCoordiante". The corresponding table is shown in Table 2.

Table 2 Coordinate value of each location

Type	Name	ID	x	y	Type	Name	ID	x	y
Rides	Concave Train	X_1	910	260	Others	Trapezoid Show	X_{11}	765	975
	Decagon Coaster	X_2	1045	390		Games	X_{12}	915	745
	Dodecagon Rapids	X_3	545	475		WC	X_{13}	1065	640
	Heptagon Coaster	X_4	660	810		Gift Shop	X_{14}	1190	1040
	Hexa-Swings	X_5	810	320		Hot Dog Stand	X_{15}	750	630
	Octagon Flume	X_6	1420	820		Ice Cream Cart	X_{16}	1295	725
	Pentagon Bumpers	X_7	560	665		Triangle Restaurant	X_{17}	1390	945
	Rhombus Ride	X_8	720	415					
	Square Scramble	X_9	1240	655					
	Star Burst Carousel	X_{10}	1140	840					

The park map including the coordinate system is shown in Figure 1.

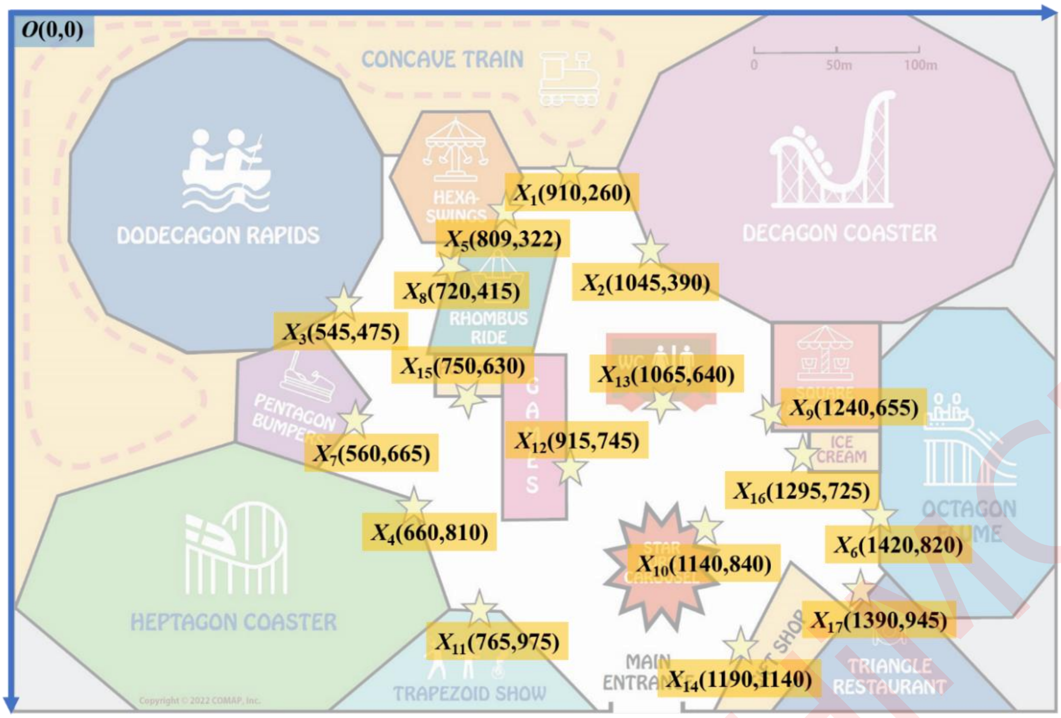


Figure 1 Park map with coordinate system

5.2 Calculate distance

Because there are few obstacles between any two locations, we define the distance between two locations as Euclidean distance [6]. we call the distance between two locations as $D_{ij}(i,j=1,2,\cdots 17)$, Its calculation formula as follow.

$$D_{ij}=\sqrt{(x_i-x_j)^2+(y_i-y_j)^2} \tag{1}$$

According to the Formula (1), we can calculate the distance between any two locations as shown in Table 3, all result is taken as an integer.

Table 3 The distance between any two locations

	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8	X_9	X_{10}	X_{11}	X_{12}	X_{13}	X_{14}	X_{15}	X_{16}	X_{17}
X_1	0	187	424	604	117	757	535	245	515	624	730	485	410	829	403	604	832
X_2	187	0	507	570	245	571	558	326	329	460	649	378	251	666	380	418	649
X_3	424	507	0	354	307	941	191	185	718	698	546	458	546	857	257	791	964
X_4	604	570	354	0	512	760	176	400	600	481	196	263	439	578	201	641	741
X_5	117	245	307	512	0	789	426	131	545	616	657	438	409	814	316	632	849
X_6	757	571	941	760	789	0	874	809	244	281	673	511	398	318	696	157	124
X_7	535	558	191	176	426	874	0	297	680	606	372	364	506	733	193	737	874
X_8	245	326	185	400	131	809	297	0	573	598	562	383	412	782	217	653	851
X_9	515	329	718	600	545	244	680	573	0	210	573	337	176	388	491	89	322
X_{10}	624	460	698	481	616	281	606	598	210	0	399	244	214	206	443	193	269
X_{11}	730	649	546	196	657	673	372	562	573	399	0	275	450	430	345	586	626
X_{12}	485	378	458	263	438	511	364	383	337	244	275	0	183	403	201	381	513
X_{13}	410	251	546	439	409	398	506	412	176	214	450	183	0	419	315	245	442
X_{14}	829	666	857	578	814	318	733	782	388	206	430	403	419	0	601	332	224

In the Table 3, The redder the color, the farther the distance, and the bluer the color, the closer the distance. The unit of Table 3 is pixel.

5.3 Calculate walking time

Now we know the distance between any two locations by pixel coordinate, and 1 centimeter in the map equal 25 meters. Let's take X_1 and X_2 as an example. The distance between X_1 and X_2 is.

$$D_{12} = 187 \text{ pixel} = 2.74\, centimeter = 68\, meter$$

According to the Centers for Disease Control and Prevention [1], we find the average walking pace is 2.5 to 4 mph, that is about 67 to 107 meter per minutes, the average value is 87 meter per minutes. Considering that our service group is students and walking in the amusement park, we take half of the average value, which is about 45 meter per minutes.

$$Tw_{ij} = \left\lfloor \frac{d_{ij}}{45} \right\rfloor \tag{2}$$

In Formula (2), Tw_{ij} means the time required to walk from location i to location j. The symbol of "$\lfloor\ \rfloor$" means round up the value to the nearest integer. we can calculate the walking time between any two locations as shown in Table 4.

Table 4 The walking time between any two locations

	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8	X_9	X_{10}	X_{11}	X_{12}	X_{13}	X_{14}	X_{15}	X_{16}	X_{17}
X_1	0	2	4	5	1	6	4	2	4	5	6	4	3	7	3	5	7
X_2	2	0	4	5	2	5	5	3	3	4	5	3	2	5	3	4	5
X_3	4	4	0	3	3	7	2	2	6	6	4	4	4	7	2	6	8
X_4	5	5	3	0	4	6	2	3	5	4	2	2	4	5	2	5	6
X_5	1	2	3	4	0	6	4	1	4	5	5	4	3	6	3	5	7
X_6	6	5	7	6	6	0	7	6	2	3	5	4	3	3	6	2	1
X_7	4	5	2	2	4	7	0	3	5	5	3	3	4	6	2	6	7
X_8	2	3	2	3	1	6	3	0	5	5	5	3	4	6	2	5	7
X_9	4	3	6	5	4	2	5	5	0	2	5	3	2	3	4	1	3
X_{10}	5	4	6	4	5	3	5	5	2	0	3	2	2	2	4	2	2
X_{11}	6	5	4	2	5	5	3	5	5	3	0	3	4	4	3	5	5
X_{12}	4	3	4	2	4	4	3	3	3	2	3	0	2	3	2	3	4
X_{13}	3	2	4	4	3	3	4	4	2	2	4	2	0	4	3	2	4
X_{14}	7	5	7	5	6	3	6	6	3	2	4	3	4	0	5	3	2
X_{15}	3	3	2	2	3	6	2	2	4	4	3	2	3	5	0	5	6
X_{16}	5	4	6	5	5	2	6	5	1	2	5	3	2	3	5	0	2
X_{17}	7	5	8	6	7	1	7	7	3	2	5	4	4	2	6	2	0

As the same, in the Table 4, The redder the color, the more time, and the bluer the color, the less time. The unit of Table 4 is minute.

5.4 Find the center location

By calculating the average time $(Tave_i, i = 1, 2, \cdots 17)$ from each location to all other locations, we find that the three least average: X_{12} **(GAMES)**, $Tave_{12} = 2.88$ minutes, X_{13} **(restrooms, WC)**, $Tave_{13} = 2.94$ minutes, X_{15} **(Hot dogs)**, $Tave_{15} = 3.24$ minutes. Obviously, the three values are very close, but X_{13} is more important than others. Calculation formula of average time as follow [7].

$$Tave_i = \frac{1}{16}\left(\sum_{j=1}^{17} Tw_{ij}\right) \qquad (i \neq j) \tag{3}$$

We choose the X_{13} as the center location, which are centrally located in the park for easy access.

5.5 The travelling salesman problem

According to the description of Wikipedia [2][5], the Travelling Salesman Problem (also called TSP problem) asks the following question: "Given a list of cities and the distances between each pair of cities, what is the shortest possible route that visits each city exactly once and returns to the origin city?"

The TSP can be formulated as an integer linear program. Common to both these formulations is that one labels the cities with the numbers $1,2,\cdots,n$ and takes $d_{ij}>0$ to be the distance from city *i* to city *j*. The main variables in the formulations are:

$$x_{ij}=\begin{cases}1 & the\ path\ gose\ from\ city\ i\ to\ j\\ 0 & otherwise\end{cases}$$

It is because these are 0/1 variables that the formulations become integer programs; all other constraints are purely linear. In particular, the objective in the program is to minimize the tour length

$$z=\sum_{i=1}^{n}\sum_{i\neq j,j=1}^{n}d_{ij}x_{ij}$$

Without further constraints, the $\{x_{ij}\}$ will however effectively range over all subsets of the set of edges, which is very far from the sets of edges in a tour, and allows for a trivial minimum where all $x_{ij}=0$. Therefore, both formulations also have the constraints that there at each vertex is exactly one incoming edge and one outgoing edge, which may be expressed as the $2n$ linear equations.

$$\sum_{i\neq j,i=1}^{n}x_{ij}=1 \text{ and } \sum_{i\neq j,j=1}^{n}x_{ij}=1$$

Our problem can also be classified as the Traveling Salesman Problem, we can use the above methods to our problems. The mathematical model is established as follows.

$$\min\ T=\sum_{i=1}^{17}\sum_{j=1}^{17}\left(Tw_{ij}\right)\times x_{ij} \tag{4}$$

$$\sum_{i=1}^{17}x_{ij}=1,\sum_{j=1}^{17}x_{ij}=1,x_{ij}=0\ or\ 1,\ for\quad i,j=1\ldots17 \tag{5}$$

In the software of “Lingo/Lindo”, we can imitate the sample description in the user’s manual. By entering the data in Table 4, we can get the result as follow, and total walking time is **31 minutes**.

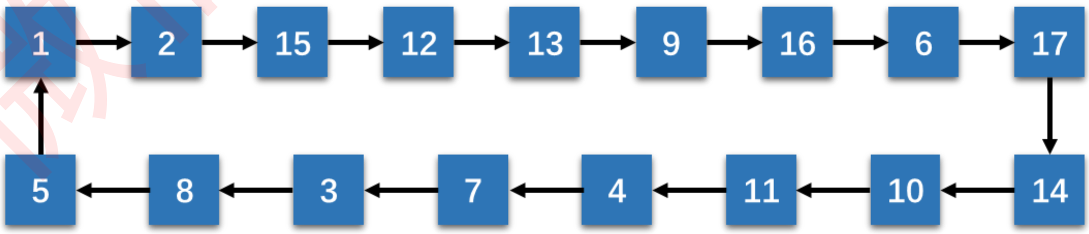


Figure 2 Shortest path scheme traversing all location

However, some locations need to be repeated and are not suitable as intermediate locations, so we choose $\boldsymbol{X_4}$ as the starting point of the trip. We draw the basic route as shown in Figure 3.

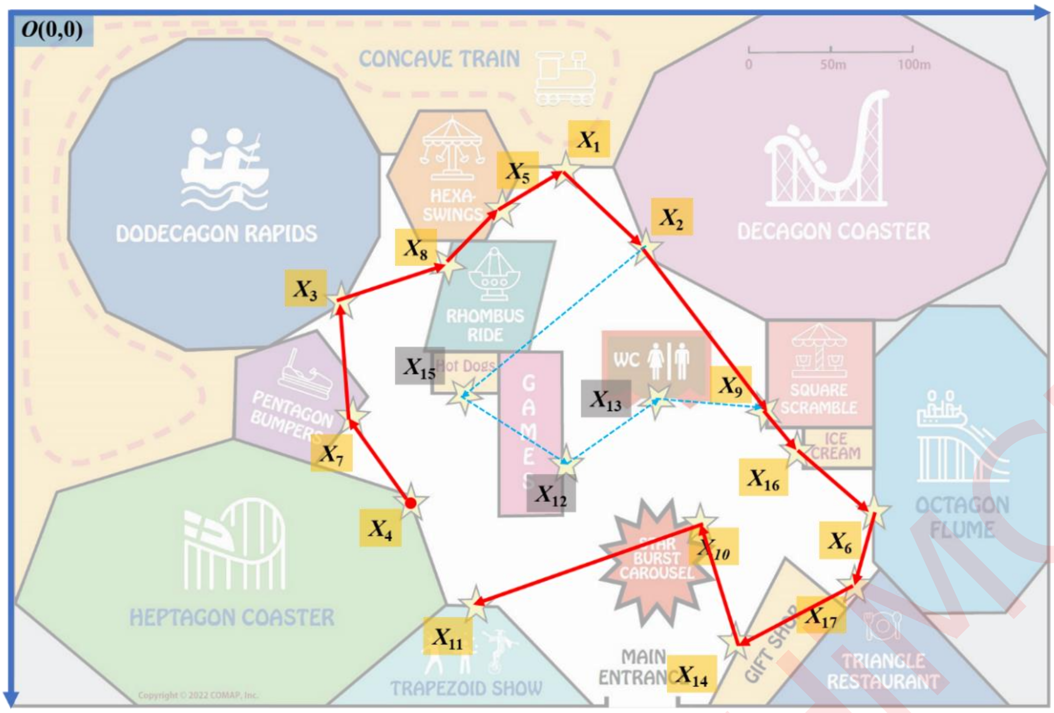


Figure 3 The basic tour map

5.6 Divide the map into several parts

Considering the dining time, the characteristics of rides and shows, the crowding degree of queues and other factors, we think it is appropriate to divide the whole park map into four parts, which are called **Part I**、**Part II**、**Part III** and **Part IV**.

We have determined that X_{13} is the center of the part. Now, we will find the farthest point from X_{13} at the top right, top left, bottom right, bottom left. They are X_2(top right), X_3(top left), X_6(bottom right) and X_{11}(bottom left). Take these points as the cluster center of each part, and properly combined with manual adjustment. The locations contained in each part and the distance between them are shown in Table 5.

Table 5 Member list of each part and the distance between them

Part I					
	X_3	X_4	X_5	X_7	X_8
X_3	0	3	3	2	2
X_4	3	0	4	2	3
X_5	3	4	0	4	1
X_7	2	2	4	0	3
X_8	2	3	1	3	0

Part II					
	X_1	X_2	X_{13}	X_{15}	
X_1	0	2	3	3	
X_2	2	0	2	3	
X_{13}	3	2	0	3	
X_{15}	3	3	3	0	

Part III					
	X_6	X_9	X_{10}	X_{16}	X_{17}
X_6	0	2	3	2	1
X_9	2	0	2	1	3
X_{10}	3	2	0	2	2
X_{16}	2	1	2	0	2
X_{17}	1	3	2	2	0

Part IV					
	X_{11}	X_{12}	X_{13}	X_{14}	X_{17}
X_{11}	0	3	4	4	5
X_{12}	3	0	2	3	4
X_{13}	4	2	0	4	4
X_{14}	4	3	4	0	2
X_{17}	5	4	4	2	0

In addition, we also counted the time-consuming distribution of partition and non-partition. As shown in Figure 4, before the partitioning, 37% of walk times were 5 minutes or more. After partitioning, the walking time of the farthest two points is 5 minutes, and only once.

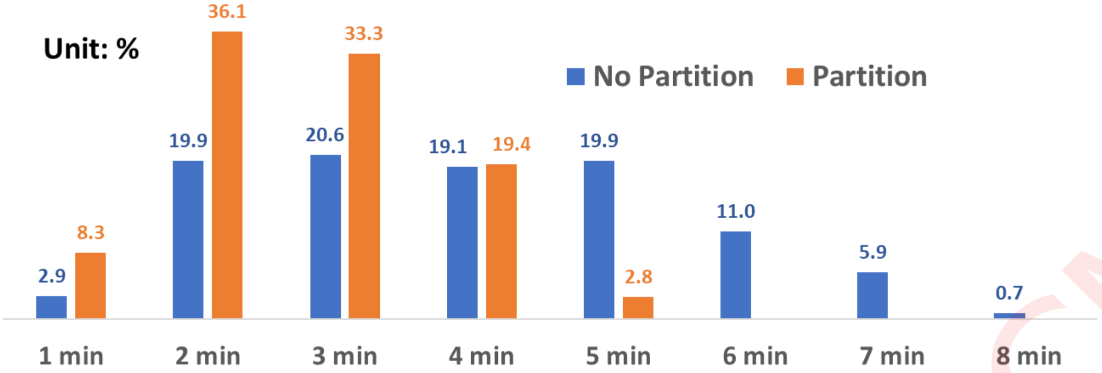


Figure 4 Walking time distribution between partitions or non-partitions

Therefore, when we formulate follow-up plans, we try to ensure that activities within a fixed period of time are only within a fixed range, which can effectively reduce walking and waiting time.

Figure 5 gives a schematic map of the 4 parts, which are distinguished by different colors. The entire play route starts from X_4 and proceeds in a clockwise direction. Some locations that need to be visited repeatedly, such as X_{13}, X_{17}, X_{11}, etc. The tour route will be adjusted appropriately according to the needs.

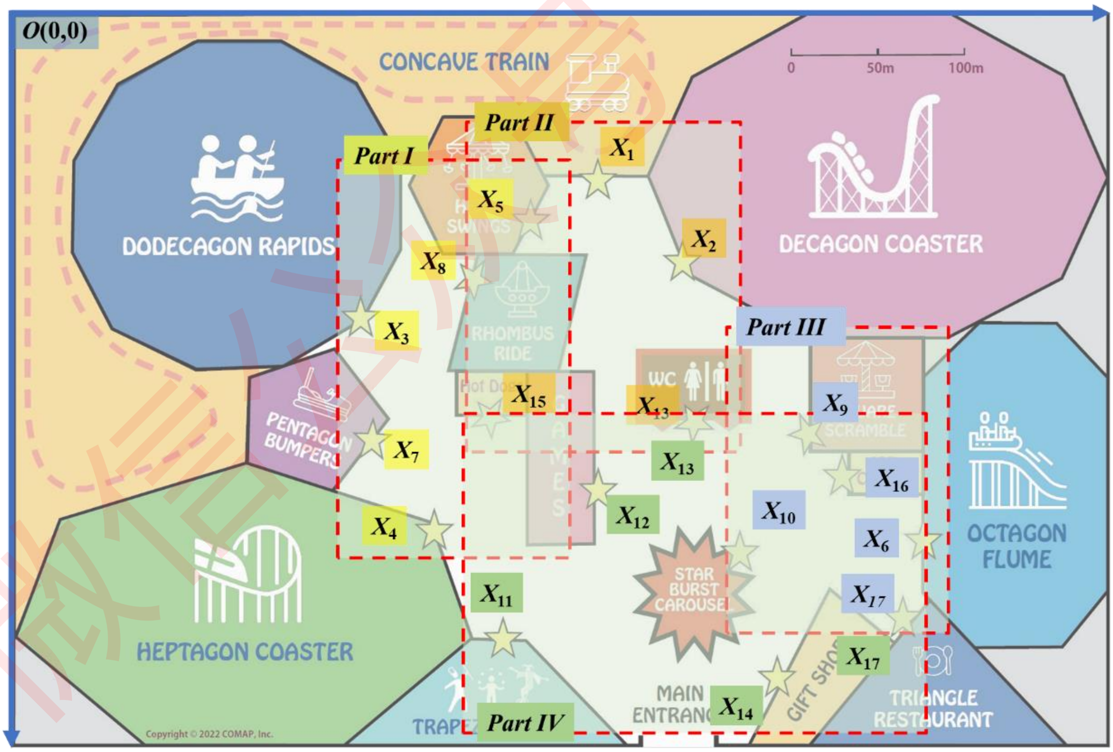


Figure 5 Schematic map of the 4 parts

6. Develop the Schedule

6.1 Arrange key locations

A total of 13 of the 17 locations have different waiting times. The heat map of the waiting time is shown in Table 6. We can easily find that there are four addresses with large waiting time peaks, which need to be dealt with first, they are X_2 (Decagon Coaster), X_4(Heptagon Coaster), X_6(Octagon Flume), X_{17}(Triangle Restaurant).

Table 6 The heat map of the waiting time

Time	X1	X2	X3	X4	X5	X6	X7	X8	X9	X10	X15	X16	X17
09:00-10:00	5	15	0	10	0	10	0	5	0	10	0	/	5
10:00-11:00	10	30	5	20	5	10	5	10	5	15	5	0	30
11:00-12:00	15	45	10	30	10	10	5	15	10	15	10	5	60
12:00-13:00	10	60	15	40	10	30	5	10	10	10	10	10	20
13:00-14:00	10	75	20	40	15	60	10	10	10	5	5	15	0
14:00-15:00	15	75	15	30	15	60	10	15	10	5	0	15	0
15:00-16:00	15	60	10	30	10	60	10	10	10	5	5	10	10
16:00-17:00	15	45	5	20	10	30	10	10	10	5	10	5	40
17:00-18:00	5	30	0	20	5	10	5	5	5	5	10	5	60
18:00-19:00	5	30	0	10	0	10	0	5	5	5	10	5	10
19:00-20:00	5	15	0	10	0	0	0	5	5	5	5	0	0
20:00-21:00	5	0	0	10	0	0	0	5	0	0	5	0	0

Follow the clockwise play route (Figure 2) and Figure 6, we decided to place the X_4(Heptagon Coaster) in 09:00-10:00, X_2 (Decagon Coaster) in 10:00-11:00, X_6(Octagon Flume) in 11:00-12:00, X_{17}(Triangle Restaurant) in 12:00-13:00 and 18:00-19:00. Total wait time for these key locations is 80 minutes, which includes the wait time for two meals [3][4][5].

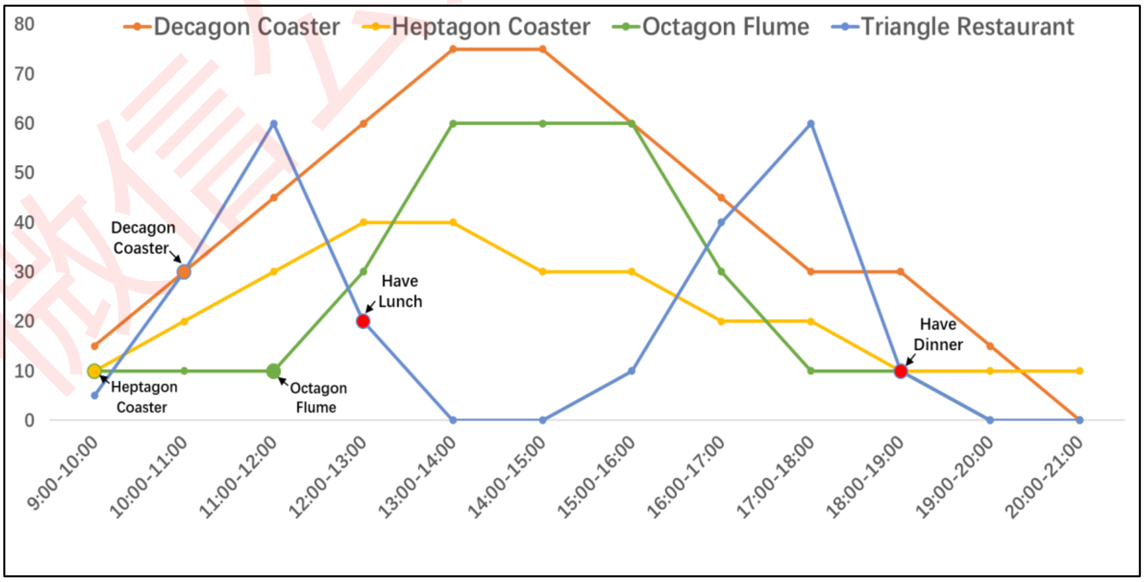


Figure 6 Line graph of waiting times at 4 key locations

6.2 Allocate time for each part

Because X_4 belongs to the first part, so the time for the **Part I** is 9:00-10:00. X_2 belongs to **part II**, the scheduled time is 10:00-11:00, so the time for the **Part II** is 10:00-11:00. X_6 and X_{17} belong to the part III, so the time for the **Part III** is 11:00-14:00. The remaining time is allocated to **part IV**. The time allocation and the location numbers included of each part are shown in Table 7.

Table7 The time allocation and the location numbers included of each part

Part I 09:00-10:00	**X3**	**Part III 11:00-14:00**	**X6**
	X4		**X9**
	X5		**X10**
	X7		**X16**
	X8		**X17**
Part II 10:00-11:00	**X1**	**Part IV 14:00-21:00**	**X11**
	X2		**X12**
	X13		**X13**
	X15		**X14**
			X17

6.3 Create the Schedule

Table 8 Schedule of the one-day trip

Part	Start Time	Location	Wait Time	Time taken	Next Location	Walking Time	End Time	Time	Tips
Part I 09:00 -10:00	9:00	X4	0:10	0:02	X7	0:02	9:14	09:00-10:00	Heptagon Coaster
	9:14	X7	0:00	0:15	X3	0:02	9:31	09:00-10:00	Pentagon Bumpers
	9:31	X3	0:00	0:04	X8	0:02	9:37	09:00-10:00	Dodecagon Rapids
	9:37	X8	0:05	0:03	X5	0:01	9:46	09:00-10:00	Rhombus Ride
	9:46	X5	0:00	0:05	X1	0:01	9:52	09:00-10:00	Hexa-Swings
Part II 10:00 -11:00	9:52	X1	0:05	0:03	X13	0:03	10:03	09:00-10:00	Concave Train
	10:03	X2	0:30	0:03	X15	0:03	10:39	10:00-11:00	Decagon Coaster
	10:39	X15	0:05	0:07	X13	0:03	10:54	10:00-11:00	Hot Dog
	10:54	X13	0:00	0:06	X9	0:02	11:02	10:00-11:00	WC
Part III 11:00 -14:00	11:02	X9	0:10	0:10	X16	0:01	11:23	11:00-12:00	Square Scramble
	11:23	X16	0:05	0:25	X6	0:02	11:55	11:00-12:00	Icecream
	11:55	X6	0:10	0:03	X17	0:01	12:09	11:00-12:00	Octagon Flume
	12:09	X17	0:30	1:00	X10	0:02	13:41	12:00-13:00	Lunch
	13:41	X10	0:05	0:05	X11	0:03	13:54	13:00-14:00	Star Burst Carousel
Part IV 14:00 -21:00	13:54	X11	0:06	1:00	X12	0:03	15:03	14:00-15:00	Magic Show
	15:03	X12	0:00	0:54	X11	0:03	16:00	15:00-16:00	Games
	16:00	X11	0:00	1:00	X14	0:05	17:05	16:00-17:00	Circus Acrobatics
	17:05	X14	0:00	0:53	X17	0:02	18:00	17:00-18:00	Gift Shop
	18:00	X17	0:10	1:00	X13	0:04	19:14	18:00-19:00	Dinner
	19:14	X13	0:00	0:42	X11	0:04	20:00	19:00-20:00	WC
	20:00	X11	0:00	1:00	/	0:00	21:00	20:00-21:00	Firework Show

6.4 Notes for each one

1.Notes to Ming

- The game time is scheduled at 15:00-16:00, about 1 hour of free time.
- All the extremely Thrill rides (X_4, X_8, X_2) are arranged before lunch, you can decide whether to participate in these rides according to your own situation.

- The waiting time of X_2 is relatively long, and the project is more exciting, you can go to Games instead. But be sure to wait by 10:39 at the hot dog stand, which is next door to the Games.

2. Notes to Ishmael

- We had nearly the entire afternoon to watch the show and could see three different shows. The time of the magic show is scheduled for 14:00-15:00 in the afternoon.
- There are three extremely thrill roller coaster rides in the park, we've all got them on the schedule.
- If you want to participate in roller coasters repeatedly, you can invite the team to play them together during the break. Available time is 17:00-18:00 or 19:00-20:00, the wait times for those times are very short. But you must not miss the time to go to the next location with the team.

3.Notes to Karine

- The good news for you is that most locations have relatively short wait times. All items except X_2 and X_{17} will have a wait time of less than 10 minutes.
- The time of the circus acrobatics show is scheduled for 16:00-17:00 in the afternoon.
- The lunch time is 12:00-13:00, and the dinner lunch is 18:00-19:00. We suggest that you can bring yourself some snacks. Except for hot dogs, there are no other snacks that can be eaten in this park.

4.Notes to Freya

- There are two water rides in the park, namely Dodecagon Rapids and Octagon Flume. Compared with other rides, their play time is relatively long, I hope you enjoy yourself.
- Our schedule allows the group to be together for most of the day. Even during free time, best friends can play some fun games into groups of two.
- The shopping time is scheduled at 17:00-18:00 at gift shop, about 1 hour of free time. Hope you find meaningful souvenirs then. In addition, bringing a camera and capturing some beautiful moments is also a good way to add meaning to the trip.

6.5 Schedule for ourself

- First of all, our team's plan will not be much different from the schedule in the Table 8.
- We will bring more food and fruit, only eat one meal in the restaurant.
- Play individual fun items in both day and night to experience different feelings.
- Actual testing on necessary data in the park to make the next travel plan more perfect.

7. Analysis

7.1 Strengths

- Arrange the right project at the right time to reduce the waiting time. According to our schedule, the average waiting time for completing all 10 rides only 7.5 minutes.
- During this day's trip, time will not waste on walking. The total walking time will be less than 1 hour. The longest walking time shall not exceed 5 minutes.
- The schedule includes enough leisure time, which can be used to have a meal, take a rest, play games or buy souvenirs.
- At the same time, we will also make targeted arrangements according to the different needs of everyone to make the collective satisfaction as high as possible.
- All rides and shows will be completed, and individual rides can be tried more than once.

The time allocation arrangement of 12 hours or 720 minutes in one day is shown in Figure 7.

Figure 7 The time allocation arrangement in one day

7.2 Weaknesses

- The schedule of some rides is too tight, and the actual situation may not be able to achieve the goal.
- The group are willing to split up into groups of two for up to 4 hours of the day if it will increase everyone's enjoyment of the trip. However, the process we used to create the schedule has not been described in detail about this point.

8. Conclusion and Advice

8.1 Our Conclusion

- Divide the whole park map into different parts is a good option.
- The optimization of the scheme can be realized through the computer-based calculation method supplemented by manual adjustment.
- Complementary issues should be considered when formulating plans for multiple teams.
-

8.2 Advice for the park

- The park should set up the restaurant in the center of the park, and increase seats to reduce waiting time.
- The overall play time of the amusement projects is too short, and the waiting time of some rides is too long, so it is necessary to properly divert and increase the number of seats for the rides.
- The ride’s name is weakly correlated with the ride’s content, and it is difficult to highlight the characteristics.

8.3 Advice for tourists

- It is recommended to bring an appropriate amount of food and beverages to solve a lunch or dinner and save time.
- You can’t give top priority to a certain project because of the short waiting time. You can make and arrange play plans in different time periods, or ask experienced friends to help make it.
- It is very necessary to combine work and rest, and it is necessary to set aside sufficient leisure time.

9. reference

[1] **The Physical Activity Guidelines for Americans**

Piercy K L, Troiano R P, Ballard R M, et al. The physical activity guidelines for Americans[J]. Jama, 2018, 320(19): 2020-2028.

[2] **Source Program of Traveling Salesman Problem**

https://www.lindo.com/doc/online_help/lingo16_0/index.html?traveling_salesman_problem.htm

[3] **Best Hollywood Studios Itinerary for One Day**

https://www.themeparkhipster.com/best-hollywood-studios-itinerary-for-one-day/

[4] **How to do Disneyland Paris in Just 1 Day**

https://www.theorganisedexplorers.com/post/how-to-do-disneyland-paris-in-just-1-day

[5] **Travelling salesman problem**

https://en.wikipedia.org/wiki/Travelling_salesman_problem

[6] **Distance Matrix**

https://en.wikipedia.org/wiki/Distance_matrix

[7] **Cluster Analysis**

https://en.wikipedia.org/wiki/Cluster_analysis